

Fields of charged bodies

W21 (2021) — English translation

Part A. Ring (2.0 points)

A ring of radius R is uniformly charged along the perimeter with a linear charge density λ .

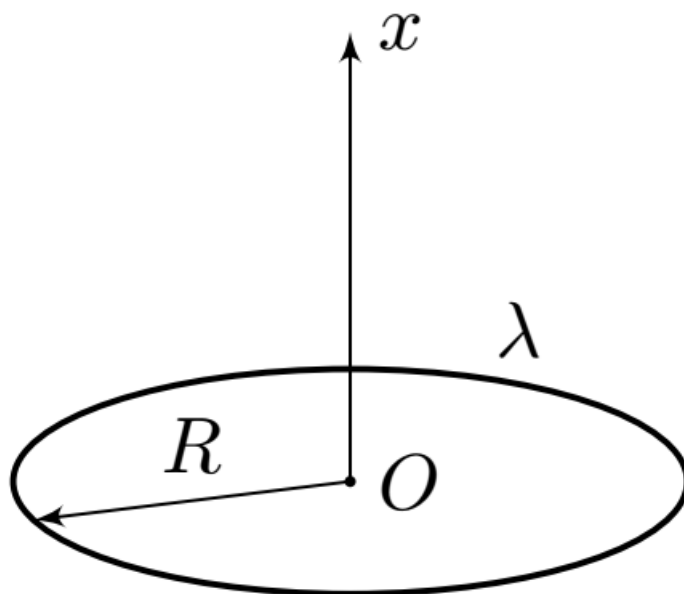


Figure 1: Figure 1.

A1	Find the field strength at the center of the charged ring.	0.20
A2	Find the potential on the axis of the ring at a distance x from its center.	0.30
A3	Find the magnitude of the field strength on the axis of the ring at a distance x from its center.	0.30
A4	At what value of x is the field strength on the ring axis maximum? Find this maximum tension.	0.60
A5	Find the field strength created by a disk of radius R , uniformly charged over the surface with charge density σ on its axis at a distance x from the center.	0.60

Part B. Two halves of the ring (2.0 points)

The ring consists of two halves of radius R , one of which is uniformly charged with linear charge density λ , and the other with $-\lambda$.

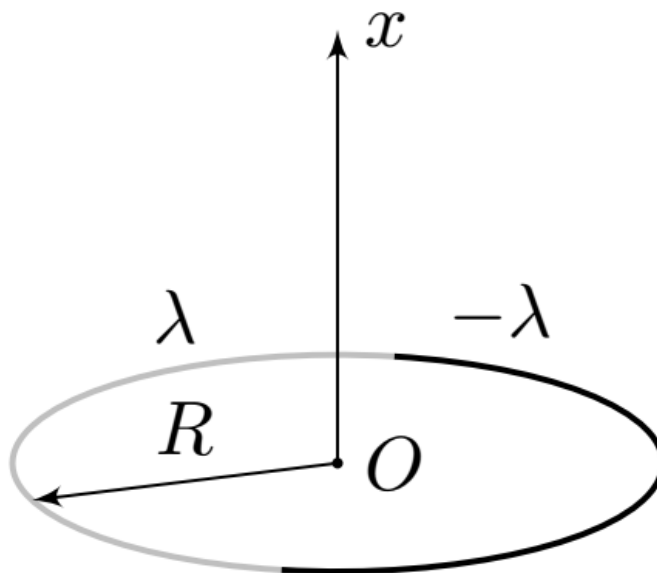


Figure 2: Figure 2.

B1	Find the potential on the axis of the ring at a distance x from its center.	0.20
B2	Find the magnitude of the field strength at the center of the ring.	1.00
B3	Find the magnitude of the field strength on the axis of the ring at a distance x from its center.	0.80

Part C. Cylinder (6 points)

An infinite cylinder of radius R is uniformly charged along the lateral surface with a surface charge density σ .

C1	What is the magnitude of the electric field strength at a distance $r < R$ from the cylinder axis?	0.30
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Now consider a semi-infinite cylinder of the same radius R , uniformly charged with the same surface charge density σ . You will need to find the field at each point at the base of the cylinder (see figure).

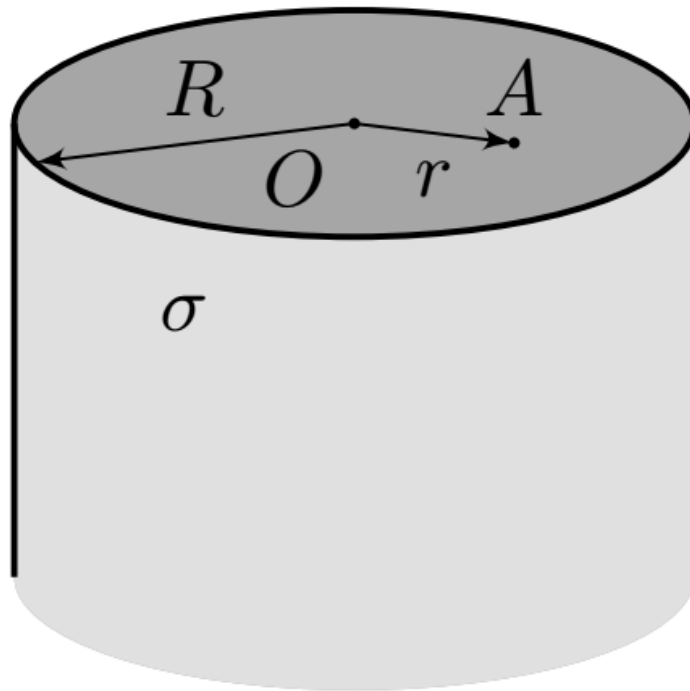


Figure 3: Figure 3.

C2	Find the modulus of the field strength at the center of the base of the cylinder O .	2.00
C3	Consider the point A at the base of the cylinder, located at a distance $r < R$ from the point O . Find the projection of the electric field strength vector onto the line OA .	0.70

To solve the next point, you may need the dependence of the radial component of the field strength of the ring in its plane. If a ring of radius R is uniformly charged with charge Q , then at a distance r from its center the radial component of the field strength is equal to

$$E_r = \frac{kQ}{2\pi R^2} \cdot y(x)$$

where $x = r/R$. The function $y(x)$ is shown in the figure below.

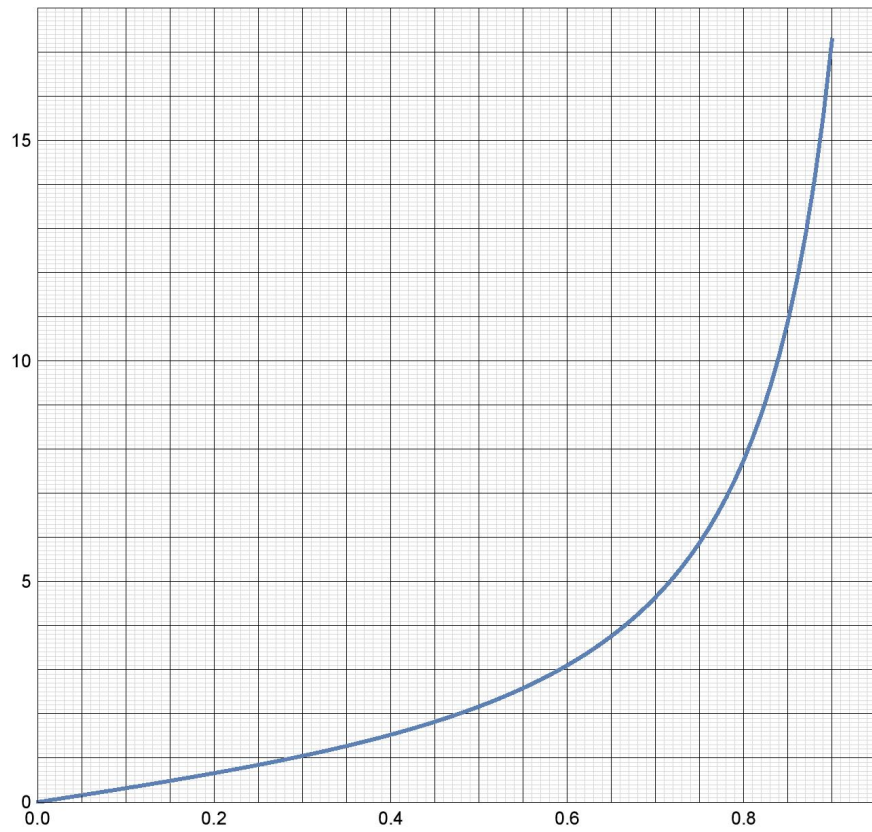


Figure 4: Figure 4.

C4 For the semi-infinite cylinder under consideration, find the magnitude of the electric field strength at point A , located at a distance $r = 0.9R$ from point O . **3.00**

Source: <https://pho.rs/p/66>